TALHA ÇAĞLAR

Computer engineering student working on cyber security, AI-assisted tools, and Linux systems.

I combine Red Team and enterprise IT internship experience with open-source applications I build end to end. This document is a curated summary of verifiable work; all source code is public on GitHub.

01

CYBER SECURITY

Authorized web testing, vulnerability analysis, OWASP Top 10, reporting.

02

AI

Applications and automations that put language models to practical use.

03

SYSTEMS

Linux desktop tools, IT infrastructure and low-level software.

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linkedin.com/in/talhacaglar1

PROFILE

EXPERIENCE AND EDUCATION

I am a final-year Computer Engineering student at Sakarya University of Applied Sciences. On the security side I have authorized testing and reporting experience; on the systems side, on-site infrastructure work and enterprise tool development.

4	5	9	2
INTERNSHIPS & PROGRAMS	CERTIFICATES & AWARDS	OPEN-SOURCE PROJECTS	AUR PACKAGES

AUG 2026 – SEP 2026
İSTANBUL · ON-SITE

PTT KULE A.Ş. IT Intern

Troubleshot physical and software faults across the large-scale IT infrastructure of Çamlıca Tower: VLAN/PoE camera-to-switch links, cabling, IP conflicts and hardware faults.

JUL 2026 – AUG 2026
BALIKESIR · ON-SITE

İŞBIR SENTETIK DOKUMA IT Intern

Built PrintHub, an Active Directory-integrated printer and toner management tool, and supported inventory and reporting workflows oriented toward ISO/IEC 27001.

DEC 2025 – MAR 2026
SAKARYA · 4 MONTHS

SIBER VATAN Student

Completed the four-month Türkiye Siber Vatan program: network and web security, vulnerability analysis and CTF-oriented exercises.

OCT 2025 – MAR 2026
REMOTE · 6 MONTHS

EMA SECURITY LTD Cyber Security Intern · Red Team

Worked on authorized web security testing, vulnerability analysis and OWASP Top 10; documented and reported findings.

EDUCATION

SAKARYA UNIVERSITY OF APPLIED SCIENCES

B.Sc. in Computer Engineering · Oct 2023 – Jun 2027 (Expected)

ACHIEVEMENT

TEKNOFEST 2024 ROBOLIG FINALIST

University level · Team AllSpark, responsible for interface design

PROJECT 01 · DESKTOP LEARNING SYSTEM

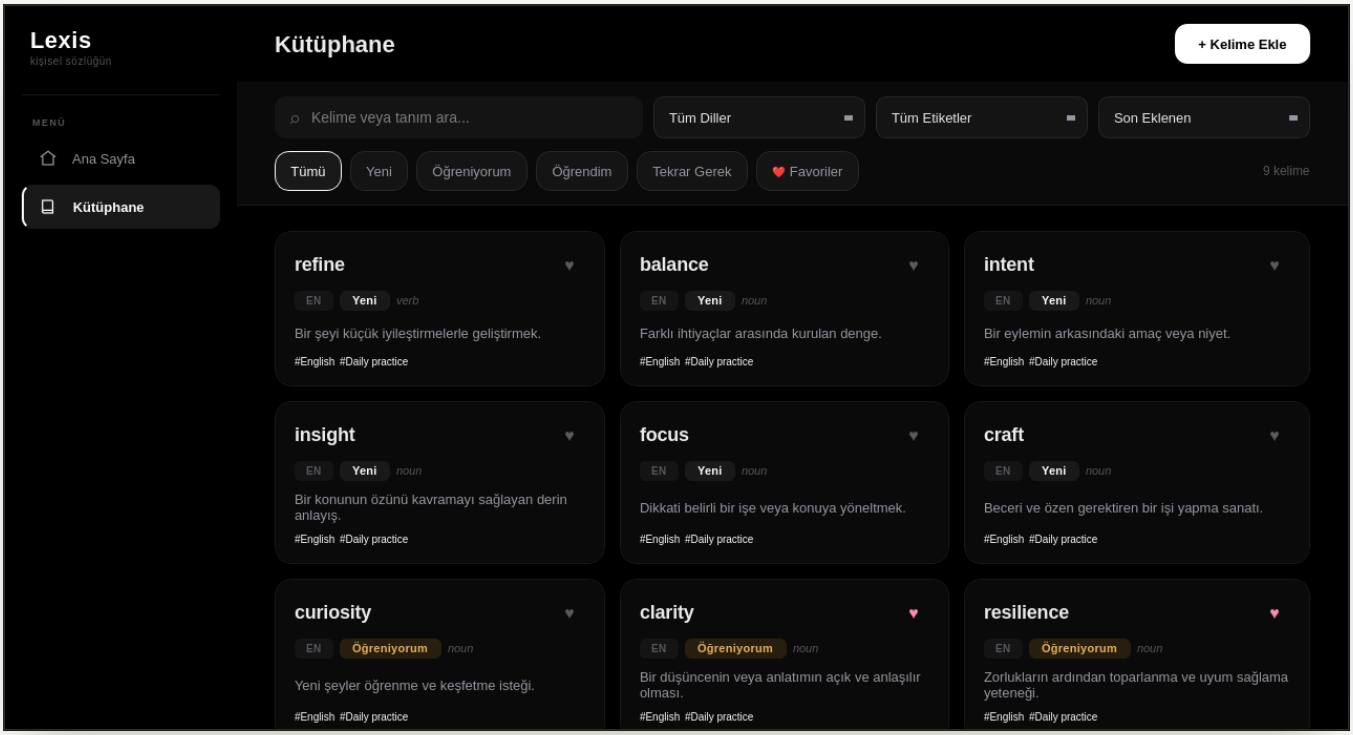
LEXIS

github.com/talhacaglar/Lexis ↗

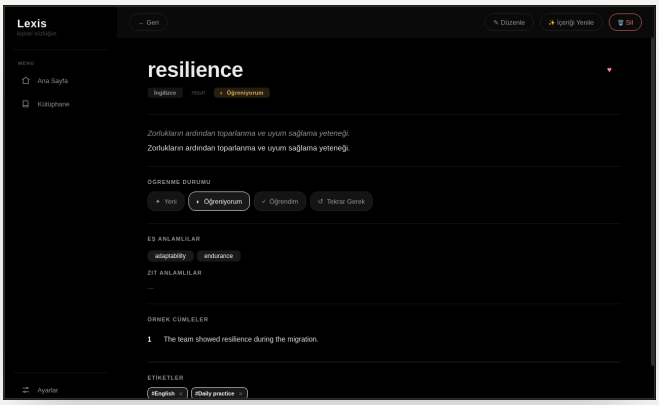
PYTHON · PYQT6 · SQLITE · GEMINI

AUR: LEXIS-GIT

A local-first, optionally AI-assisted dictionary and language-learning app. Context, tone, examples, Turkish translations and spaced repetition in one desktop workflow.



Library: search, tags, learning states



Word detail: definitions, examples, synonyms

- No API key: Wiktionary, dictionaryapi.dev and Tatoeba
- SM-2 spaced repetition, streaks and a 7-day activity graph
- All data stays local in ~/.lexis/lexis.db
- Gemini assistance only when the user opts in

PROBLEM	DECISION	RESULT
Useful dictionary tools for Turkish learners are often subscription-based or locked behind API keys.	Combine open dictionaries with optional Gemini assistance and keep every saved word in local SQLite.	A keyless desktop workflow with SM-2 practice, import/export and an AUR package.

PROJECT 02 · FOCUS INFRASTRUCTURE

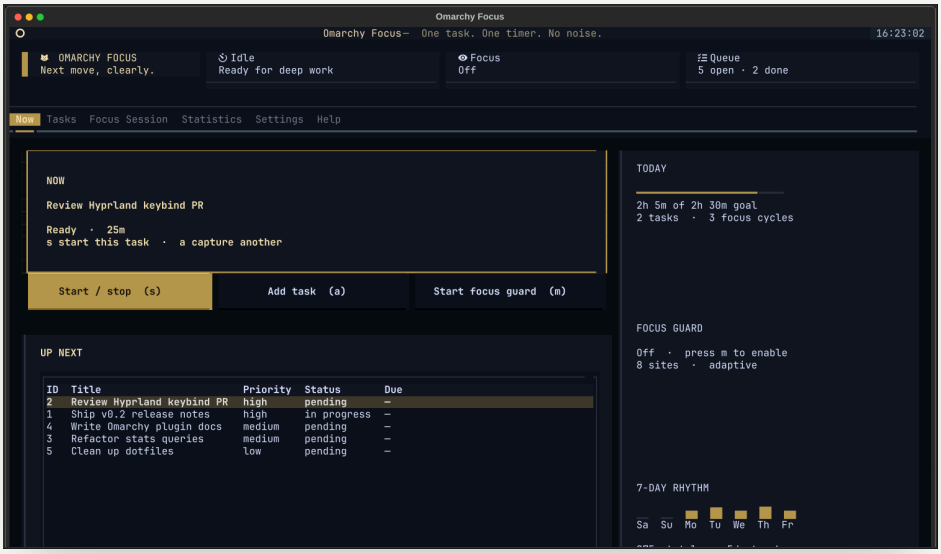
OMARCHY FOCUS

github.com/talhacaglar/omarchy-focus ↗

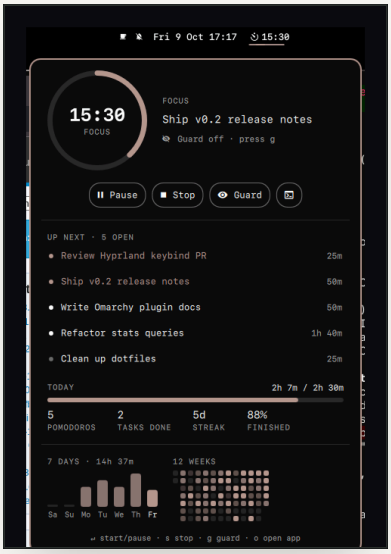
PYTHON · TEXTUAL · QML · SQLITE

AUR: OMARCHY-FOCUS

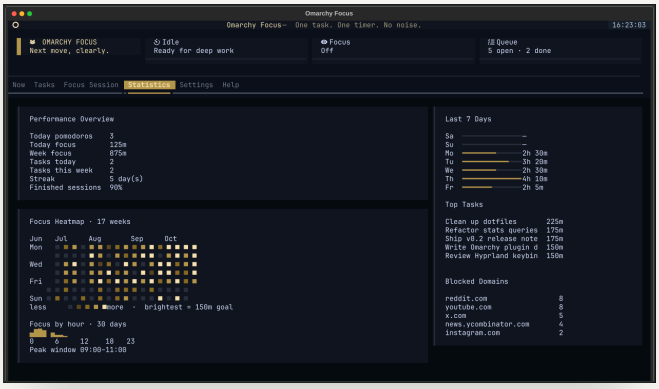
Pomodoro timer, task queue and a reversible site blocker for Omarchy: a native bar widget, a Textual terminal app and a CLI sharing one local SQLite database.



Terminal app: Now view and task queue



Bar panel



Statistics: goal, streaks and heatmap

- No polling: the widget watches a change file with inotify
- Elevation through the polkit dialog; strict mode and recovery
- Daily goal, streaks and a GitHub-style heatmap
- Keyboard-first TUI and a scriptable CLI

PROBLEM	DECISION	RESULT
Tasks, focus timers and distraction blocking were split across unrelated tools and fragile shell state.	A SQLite service layer and marked, reversible /etc/hosts blocks with crash recovery.	A recoverable focus system with a live bar widget, heatmap statistics and an AUR package.

PROJECT 03 · IT OPERATIONS CONSOLE

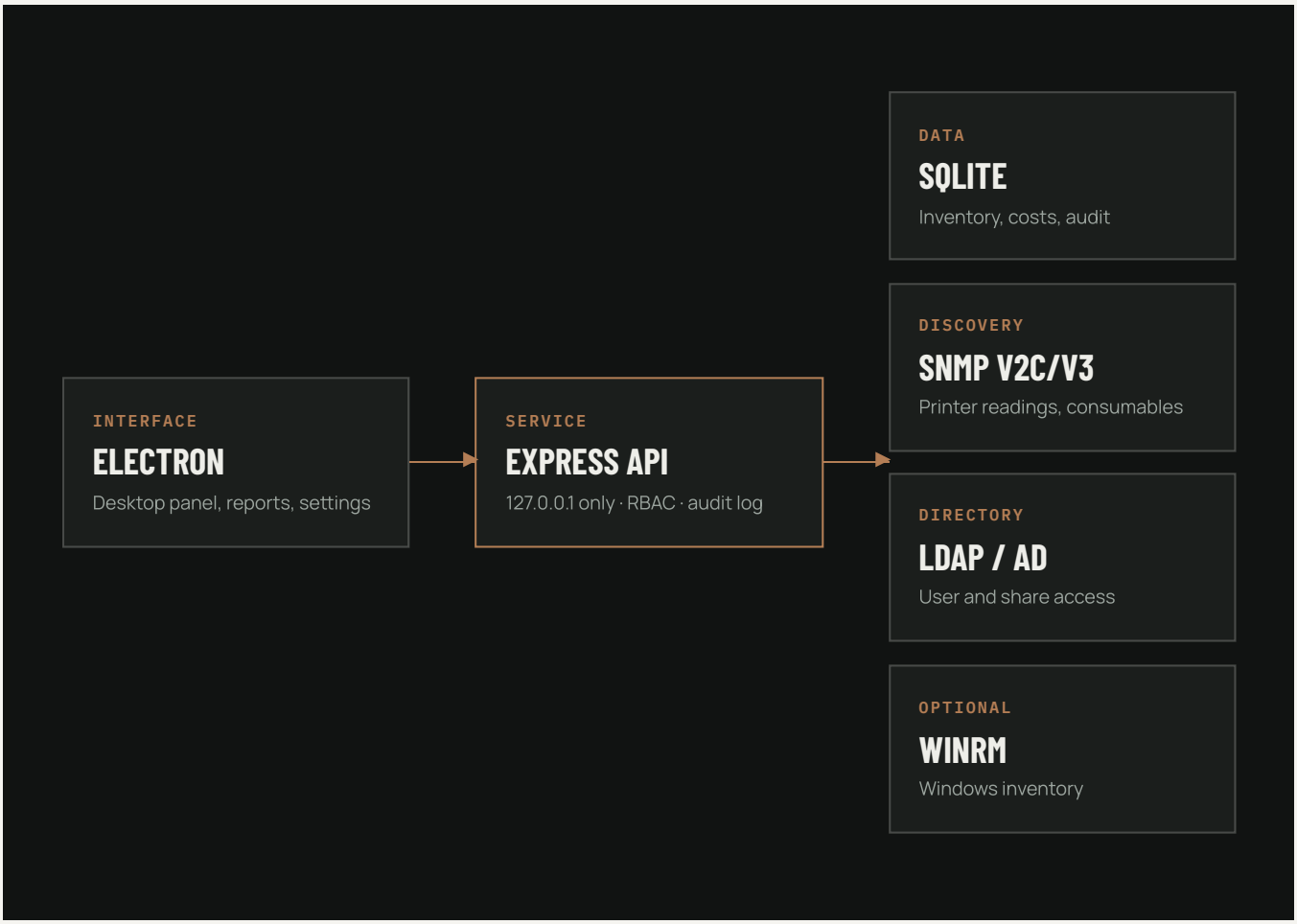
PRINTHUB

github.com/talhacaglar/PrintHub ↗

ELECTRON · EXPRESS · SQLITE

BUILT DURING THE İŞBIR INTERNSHIP

A desktop workspace for network printer monitoring, toner stock and costs, and Active Directory operations. The backend listens only on 127.0.0.1.



PROBLEM

Printer health, toner stock, costs, directory access and audit evidence lived in separate operational views.

DECISION

Unify SNMP monitoring, LDAP access, inventory, RBAC and audit records in a local Electron application.

RESULT

One panel for day-to-day printer operations with CSV and ISO/IEC 27001-oriented reporting.

- Mandatory password change on first setup
- Separate Samba AD test environment documented
- Role-based permissions and audit records
- Automated tests in the repository (npm test)

ELECTRON · EXPRESS · SQLITE · SNMP · LDAP · ACTIVE DIRECTORY · RBAC · ISO/IEC 27001

SECURITY WORK AND CREDENTIALS

My security experience rests on authorized testing, structured training programs and verifiable certifications.

INTERNSHIP

EMA SECURITY LTD

OCT 2025 – MAR 2026 · REMOTE

Six-month Red Team internship: authorized web security testing, vulnerability analysis and OWASP Top 10. Validating and documenting findings and communicating technical observations clearly.

PROGRAM

TÜRKIYE SIBER VATAN

DEC 2025 – MAR 2026 · SAKARYA

Four-month program: hands-on, CTF-oriented work in network and web security, vulnerability analysis and security fundamentals.



OCT 2024

TEKNOFEST 2024 ROBOLIG FINALIST

Reached the final of the TEKNOFEST Robolig Competition (university level and above) with Team AllSpark; I was responsible for interface design within the team.

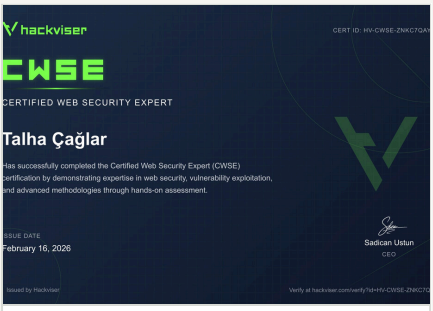
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JUN 2026

TÜRKIYE SIBER VATAN PROGRAM 2026

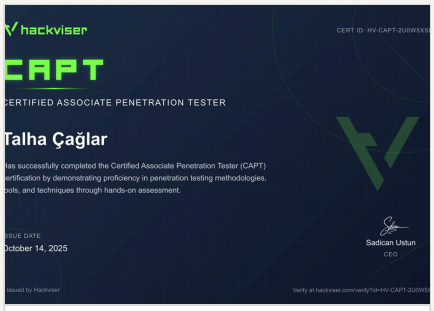
Certificate of Participation · 32735234861993



FEB 2026

CERTIFIED WEB SECURITY EXPERT

Hackviser · HV-CWSE-ZNKC7QAY



OCT 2025

CERTIFIED ASSOCIATE PENETRATION TESTER

Hackviser · HV-CAPT-2U0W5XSE



AUG 2025

NETWORKING BASICS

Cisco Networking Academy

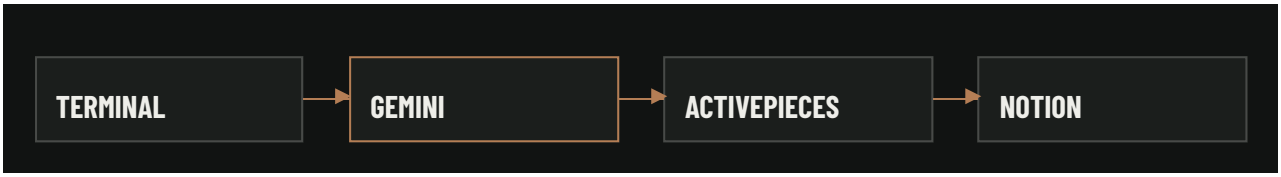
MORE PROJECTS

TOOLS AND EXPERIMENTS

AI AUTOMATION

NOTION DICTIONARY

Requests structured JSON (definition and example sentence) from Gemini for a word typed in the terminal and delivers it to Notion through an ActivePieces webhook.



PYTHON · GEMINI API · ACTIVEPIECES · NOTION

LOW-LEVEL TOOLCHAIN

RV32I ASSEMBLER & LINKER

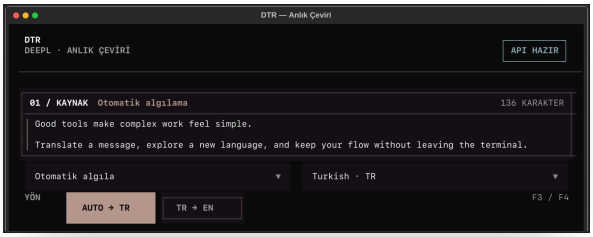
RV32I assembler and linker in C: object files, symbol resolution, relocations, configurable memory regions and PicoRV32-oriented output, with a Flask interface.

C · RISC-V · FLASK · PICORV32

TERMINAL UTILITY

DTR · INSTANT TRANSLATOR

A small DeepL-powered terminal interface for translating without leaving the shell.



PYTHON · TEXTUAL · DEEPL

SYSTEM MAINTENANCE

ARCHSWEEP

Dependency-free, opt-in Arch Linux maintenance TUI: orphans, pacman/AUR caches and journal cleanup. Nothing runs until you select it.

BASH · ARCH LINUX · TUI

MOBILE APP

POCKETFARM

Android marketplace connecting farmers and buyers directly: map-based listings, real-time chat and per-kilogram offers.

JAVA · ANDROID · FIREBASE · GOOGLE MAPS

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